



Effective control of blood glucose status and toxicity in streptozotocin-induced diabetic rats by orally administration of vanadate in an herbal decoction

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ABSTRACT

Vanadium compounds have been well recognized for hypoglycemic effects, but questions remain on gastrointestinal disturbance and possible tissue vanadium accumulation thus slowing the acceptance of vanadium compounds as diabetic therapeutic agents. Our intestinal permeability and toxicity studies of vanadium compounds have suggested that the co-administration of vanadate with *Salvia miltiorrhiza* Bunge decoction could benefit the therapeutic use of hypoglycemic vanadium compounds. In the present paper, we tested the hypoglycemic effects of vanadate ingested in an aqueous extract of *S. Bunge* using a streptozotocin (STZ)-induced diabetic rat model. Oral administration of vanadate in *S. Bunge* herbal decoction produced a stable (free of hypoglycemic shock) and long-lasting (~70 day) control of blood glucose status. Effective protection of animal organs from hyperglycemic damage was also observed. As expected, the herbal extract significantly alleviated vanadium toxicity, i.e. GI stress and metal accumulation. In addition, the result suggesting that vanadium-induced amelioration of the diabetic state appears to be secondary to the preservation of a functional portion of the pancreatic β -cells which initially survived STZ-toxicity. These studies provide new insight into the therapeutic treatment of diabetics with vanadium compounds.

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1. Introduction

Diabetes mellitus is a metabolic disease featured by increased circulating glucose level (Rosenbloom, 2007). Many metabolic problems are associated with diabetes such as cardiovascular and renal disease as well as ocular and nervous system complications (Wan et al., 2007). Diabetic patients have a 3-fold higher rate of developing cardiovascular disease, the leading cause of mortality for diabetics. Moreover, early onset of cataract and kidney failure significantly influence both morbidity and mortality rate (Santoso, 2006).

Vanadium compounds have been recognized as potent hypoglycemic agents (Shechter et al., 2003). In 2000, a vanadium-based candidate drug (bis(ethylmaltolato)oxovanadium, BEOV) was in Phase I clinical trial (Thompson and Orvig, 2006). Promising results were obtained, “but not without some difficult remaining ques-

Abbreviations: BEOV, Bis(ethylmaltolato)oxovanadium(IV); *S. Bunge*, *salvia miltiorrhiza bunge*; OGTT, oral glucose tolerance test; AKP, alkaline phosphatase; ALT, alanine aminotransferase; LDH, lactate dehydrogenase; TC, total cholesterol; TG, triglyceride; HDL-C, high density lipoprotein cholesterol; LDL-C, low density lipoprotein cholesterol.

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tions” (Thompson and Orvig, 2006) including the clarification of the vanadium hypoglycemic mechanism. The adverse side effects of vanadium treatment in diabetic animal models and human studies have slowed the acceptance of these agents as diabetic medications (Poucheret et al., 1998). Gastrointestinal disturbance has been a major short-term toxic effect of vanadium administration, reducing the food intake in treated animals and/or causing severe diarrhea that lead to animal mortality (Mukherjee et al., 2004). The long-term toxicity concern was possible vanadium accumulation in some important tissues (bone > kidney > liver) (Thompson and Orvig, 2006). Other toxic effects of vanadium included deaths, weight loss, genotoxicity, changes in body functional markers (e.g. decrease of organ-body weight ratios, increase of blood urea and creatinine levels, and lipid peroxidation), and potential carcinogenicity (Domingo, 2000). Recent studies have indicated that vanadium-mediated generation of reactive oxygen species (ROS) play an important role in vanadium-induced toxicity (Shukla et al., 2006). The biochemical mechanisms of toxicity and carcinogenicity induced by vanadium arise from elevation of reactive oxygen species such as O_2^- and H_2O_2 leading to activation of the transcription factor activator protein-1 (Shukla et al., 2006).

Recently, the intestinal permeability and toxicity of vanadium compounds demonstrated that vanadium compounds could be absorbed by forming ion pair complexes (Zhang et al., 2006). Thus

providing alternative way of improving the oral absorption of anionic vanadium compounds, such as vanadate. The toxicity of vanadium compounds has been shown to be closely associated with reactive oxygen species (ROS) induction and both enzymatic and non-enzymatic antioxidants that could reduce vanadium-induced oxidative stress (Shi et al., 1996). To reduce the vanadium toxicity, vanadate salts were ingested with botanic antioxidants such as being suspended in a lichee black tea decoction (Clark et al., 2004a) and (Clark et al., 2004b) or dissolved in green tea (Edel et al., 2006) and (Soussi et al., 2006). Co-administration with botanic antioxidants produced a significant reduction in vanadate toxic side effects and decreased vanadium accumulation in diabetic rats (Soussi et al., 2006).

Salvia miltiorrhiza Bunge is an herbal preparation widely used in Traditional Chinese Medicine for treatment of cerebrovascular disorders, angina pectoris, hypertension, chronic hepatitis, and cancer (Chen and Lin, 1999). *S. Bunge* contains a number of components with excellent antioxidative activity, including: tanshinones and tanshinols that prevent lipid peroxidation (Niu et al., 2000); Salvianolic acid A and B and rosmarinic acid that inhibit Fe(II)-induced lipid peroxidation in microsomes and prevent superoxide formation in xanthine–xanthine oxidase system (Huang and Zhang, 1992); and Lithospermate B a hydroxyl radical scavenger (Yokozawa et al., 1995). *S. Bunge* can also reduce oxidative stress *in vivo* by inducing antioxidant enzymes. For example, tanshinone II-A can increase the activity of superoxide dismutase, glutathione peroxidase, and catalase in mouse hearts, thus protecting Adriamycin-induced lipid peroxidation (Forge and Schacht, 2000). The extraordinary antioxidant capacity of *S. Bunge* provides the potential to control vanadium toxicity.

Absorption, excretion and storage mechanisms of vanadium in living system are not thoroughly understood. Several reports show that vanadium is poorly (only about 10%) absorbed from gastrointestinal tract (Mukherjee et al., 2004). Unabsorbed vanadium is excreted in feces. Vanadium has been reported to be excreted through bile and through urine. It is, thus, the bile route through which a significant amount of vanadium may ultimately excreted through feces. Those absorbed vanadium after reaching the blood stream are rapidly transported by blood proteins (transferrin and albumin) to various tissues. Plasma vanadium concentrations of those treated rats were significantly higher when sodium orthovanadate suspended in a Lichee Black Tea compared with that suspended in a double distilled water (Clark et al., 2004a). Reports suggests that use of eligible preparator can increase vanadium absorption and decrease its toxicity.

In the preliminary studies (S1), we observed that a water extract of *S. Bunge* improved the apparent permeability coefficient (P_{app}) of sodium vanadate (NaVO_3^-) across a caco-2 model (a recognized model for estimating intestinal permeability) from 8×10^{-7} cm/s to 3×10^{-5} cm/s. The *S. Bunge* extraction almost eliminated vanadium-induced damage to the cell tight junction, which is a sensitive marker for vanadium toxicity (Zhang et al., 2006). These results suggested co-administration of *S. Bunge* decoction could benefit the therapeutic application of hypoglycemic vanadium compounds. In the present paper, we tested the hypoglycemic effects of vanadate alone and when ingested in the aqueous extract of *S. Bunge* in a STZ-induced diabetic rat model.

2. Research design and methods

2.1. Materials and animal model

All the reagents and biochemicals, unless otherwise mentioned, were obtained from Sigma Chemicals Co. (St. Louis, MO, USA) and Roche Inc. (Germany).

Adult male Wistar rats weighing 200 ± 20 g at the beginning of the experiments were from the experimental animal center of Shandong University (SCXK 2005 0017). The animals were acclimatized to standard laboratory conditions (temperature

24 ± 2 °C, relative humidity $55 \pm 5\%$ and a 12-h photoperiod) in polycarbonate cages (four rats per cage) for two weeks before the experiment. During the entire study period, the rats offered free access to both food (standard laboratory diet GB14924.3-2001) and water. The animals were starved overnight with free access to water before any experiments. The animal experiments were conducted in accordance with the Guidelines for the Care and Use of Laboratory Animals.

Diabetic rats were induced by a single intravenous injection of streptozotocin (STZ) in pH 4.5 citrate buffer at a dosage of 40 mg/kg body weight via the tail vein. Control animals received an injection of the citrate buffer. The plasma glucose levels of rats were measured 7 days after injection. The diabetic animals chosen for the study had plasma glucose of 13.9 mmol/L or higher.

2.2. Treatment of diabetes

The vanadium-herb decoctions were prepared by dissolving 15 mg sodium metavanadate in 1.0 mL *S. Bunge* decoction. The herb decoction was prepared by soaking 100 g leaves of *S. Bunge* (Laiwu mountain miao herb plant depot, Shandong, China) in 1.0 L hot de-ionized water at 90 °C for 15 min in the dark with frequent stirring. The extract was cooled for 1 h before filtering through 2 layers of cheesecloth. The filtered solution was re-filtered through 4 layers of cheesecloth. The vanadium-herb decoction was made with 15 mg of sodium metavanadate added to each 1-mL aliquot of herbal decoction. The sodium metavanadate was suspended with stirring into herbal decoction. The solution was then stored for 1 h in the dark at room temperature until used. Unused vanadate-herb decoction was discarded. Fresh solution was made daily. In order to properly compare the results of the vanadate-herb decoction with the conventional vanadate preparations used in the past, a vanadate-water solution was made in an identical fashion (except without the herb) and also given to the diabetic animals. Sodium metavanadate at 15 mg/mL was suspended in de-ionized water and stored at room temperature in the dark for 2 h prior to animal treatment.

The treatment of animal groups are summarized in Table 1. At 10:00 h, animals with a daily blood glucose level ≥ 10 mmol/L was orally gavaged with 2 mL of one of the two vanadium solutions. Rats with daily blood glucose levels < 10 mmol/L were considered normoglycemic and not treated on that day. Daily per cage food and water consumption were recorded at 08:00 h. Animal body weights of each group were measured weekly.

2.3. Blood glucose analysis and biochemical assays

Daily blood glucose levels were measured using the glucose test strips with a Glucometer Elite system (Roche Diagnostics, Germany) according to the manufacturer's protocol. In the first test, 5 μ L of blood was collected with a distal tail snip. Later, the tail scab was removed at 09:00 h to allow for daily glucose testing.

Plasma insulin concentrations were determined using a radio immunological kit (Phadaseph Insulin RIA, 3 V company, China) using the method of the manufacturer. Plasma alanine aminotransferase (ALT), lactate dehydrogenase (LDH), alkaline phosphatase (AKP), urea nitrogen, creatinine, total cholesterol (TC), triglyceride (TG), high density lipoprotein cholesterol (HDL-C), and low density lipoprotein cholesterol (LDL-C) were measured with diagnostic kits (Hitachi 7060 type, Japan) according to the protocols of the manufacturer.

2.4. Tissue samples and vanadium determination

Finally, the rats were anesthetized using sodium pentobarbital (50 mg/kg, i.p.) and sacrificed. Tissue samples of liver, heart, pancreas, spleen, kidney, lung, epididymis fat pad, testes, brain, and skeletal muscle were removed immediately. After washing with saline, the tissue samples were blotted dry and weighed. All samples were stored at -80 °C for future analysis.

Weighed aliquots of frozen tissue were digested in 3 stage: the first using 5 mL mixed acids (4:1 nitric acid: perchloric acid), the second using a combination of 2 mL HNO_3 and 30% H_2O_2 , and finally using 2 mL HNO_3 . All digestions were performed at 130 °C until the acid was completely evaporated and the residue dried

Table 1
Groups of the animals and methods of treatment

Group label	Animal	Animals numbers	Daily treatment method
D	Diabetes rats, control	8	Free access to standard rat diet and water
V	Diabetes rats, treated with vanadate	8	Gavaged 2 mL 15 mg/mL vanadate-water solution, and Free access to standard rat diet and water
HV	Diabetes rats, treated with vanadate-herb decoction	8	Gavaged 2 mL 15 mg/mL vanadate-herb decoction, and free access to standard rat diet and water
C	The control normal rat	4	Free access to standard rat diet and water

before the next acid stage was started. After the third acid treatment, 1% HNO₃ was added to the digests and heated at 80 °C for 1 h. After cooling, the sample volume was measured and analyzed. Sample specimens were diluted as necessary. Standard solutions of vanadium were prepared from a certified reference standard of vanadium (1 mg/mL, mediator 10% HNO₃, GSB G 62016-90).

Vanadium concentrations were determined using atomic absorption (AA) equipped with a polarized Zeeman graphite furnace spectrophotometer. The detection wavelength was 318.4 nm with a slit width of 0.40 nm. The instrumental program included: drying at 80 °C to 140 °C ramped over 40 s; incineration at 900 °C for 20 s; and atomization at 2700 °C for 7 s. At the end of the assay, the furnace was cleaned by heating at 2800 °C for 7 s. Vanadium concentrations were calculated from the standard curve (linear range: 2 ng/L to 150 ng/L of vanadium, $r = 0.9996$). All assays were done in duplicate.

2.5. Oral glucose tolerance test (OGTT) procedure

The rats were anesthetized with an intraperitoneal (i.p.) injection of 40 mg/kg sodium pentobarbital. The right arteria saphena was exposed and isolated through a middle inguina incision. Thin silastic catheter was introduced and secured with a silk ligature. The free end of the catheter was attached to short segments of steel tubing and tunneled subcutaneously through the back to the top of the skull and through a skin incision to the exterior. After patency had been checked, the catheter was flushed with 500 μ L 0.9% NaCl containing heparin (200 IU/mL), and then 500 μ L sodium penicillin G (10,000 U/mL) was injected. The catheter was then filled with a viscous solution of 50% polyvinylpyrrolidone (PVP-30) and capped. Sodium penicillin G (10,000 U/mL) was injected every day to prevent infection.

The rats were allowed to recover from surgery for at least 4 days. The OGTT studies were done in the morning after a 16 h overnight fast. Throughout the study, rats were awake and allowed to move freely within a large cage. The PVP solution was aspirated from the catheter, and the 0-h blood samples were drawn from the arterial cannula for glucose (5 μ L per sample), insulin (0.5 mL), and biochemistry index (3 mL) determinations. All the blood samples except those for glucose tests were collected in Eppendorf tubes pre-rinsed with heparin solution (20 IU/mL) and kept in ice. The plasma was separated by centrifugation (3500g at 4 °C for 20 min). The plasma specimens for the biochemistry index and insulin assays were preserved at 4 °C and –70 °C, respectively. Then a glucose solution (1 g/mL), was immediately given to the animal at a dose of 2 g/kg body weight by means of gavage. After glucose administration, blood samples were withdrawn for the listed tests after 30, 60, 90 and 120 min.

2.6. Statistical analysis

Statistical treatment of data was performed using an analysis of variance (ANOVA) test followed by a Duncan's post hoc test. Results were reported as the mean $\pm$ SD. Statistical significance was determined at $P < 0.05$.

3. Results

3.1. The hypoglycemic effects of vanadium salts

The blood glucose levels of the animals were monitored and the results are shown in Fig. 1. The C rats maintained an average glucose level of ~ 5 mmol/L during the experimental period. The D rats had an average glucose level of ~ 20 mmol/L in the beginning (7 days after STZ injection), but their blood glucose concentration kept rising to 32 mmol/L during the first month, then decreased and stayed at ~ 27 mmol/L. Vanadate was administrated by gavage either dissolved in de-ionized water (V) or the herbal decoction (HV). The treatment plan was to withhold an additional dose of drug as long as the blood glucose level was < 10 mM. The administration of vanadate in de-ionized water at a dose of 30 mg/day/animal resulted in rapid decrease of blood glucose levels within 24 h, but the glucose level went up on the second day. The continual treatment of diabetic rats with vanadate-water solution reduced the blood glucose to the control level within one week. Then the average glucose level of 8.6 mM fluctuated and frequently increased so that the regular dose of vanadate was necessary to maintain the control blood glucose level. Interestingly, the blood glucose level of the diabetic rats given the vanadate-herb decoction continually decreased to the control level in ~ 5 days. Then, the glucose level stayed at an average level of ~ 7.0 mM for the next two months; an average of 71 ± 3 consecutive days for a blood glucose level of < 10 mmol/L. Then, the glucose level was variable, so additional drug was given.

3.2. Effect of vanadium on the insulin levels of animals

Fig. 2 shows the results of the basal fasting blood glucose levels (Fig. 2a), plasma insulin levels (Fig. 2b) in the oral glucose tolerance test at the end of the experiment (15 weeks). There was no significant difference in the basal fasting blood glucose levels, but the D rats exhibited a much lower level of basal fasting insulin than the test rats. During the 2 h following glucose ingestion, all the treated

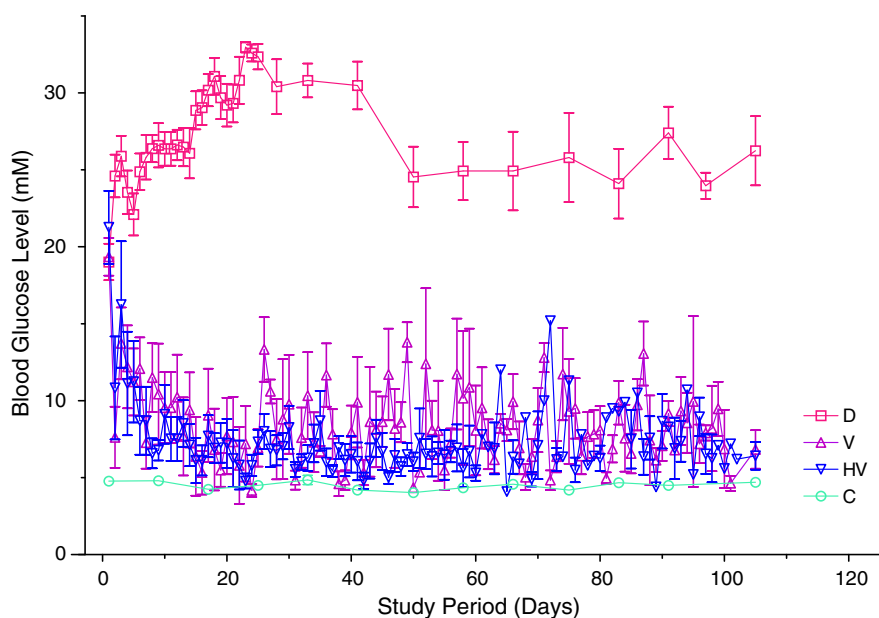


Fig. 1. Daily blood glucose levels of diabetic rats with vanadium treatment. The D rats were untreated diabetic, V rats were treated with vanadate-water solution, HV rats were treated with vanadate-herbal decoction, and C rats were non-treatment control normal rats. Data of the survivors were represented as mean $\pm$ SD with $n = 7, 6, 8$, and 4 for D, V, HV, and C rats, respectively.

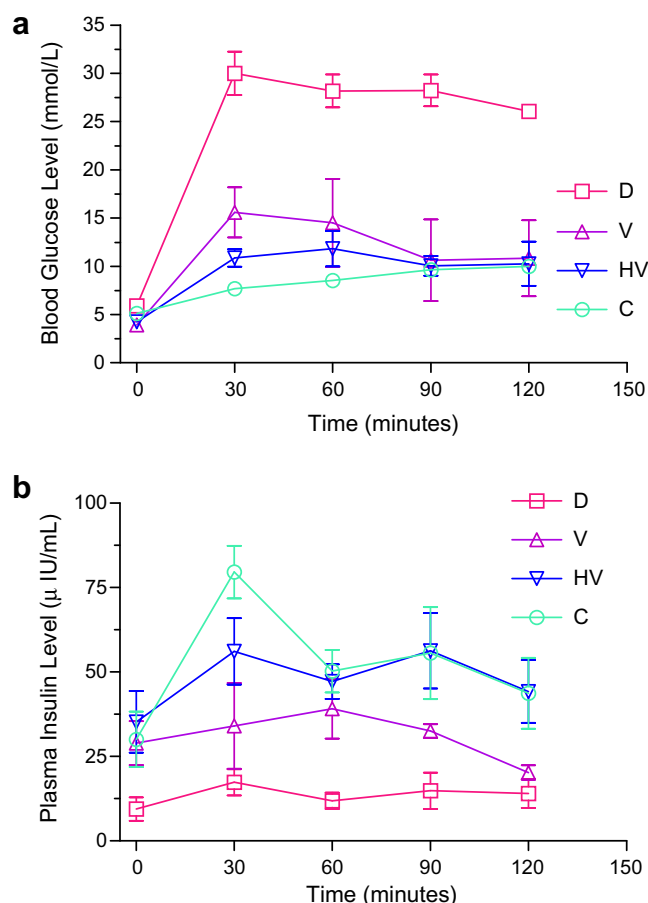


Fig. 2. Changes in blood glucose (a) and plasma insulin (b) in an oral glucose tolerance test. Rats were given 2 g glucose/kg BW in the tests performed after 15 weeks. Data of the survivors represented as the mean $\pm$ SD with $n = 7, 6, 8$, and 4 for D, V, HV, and C rats, respectively.

rats exhibited smaller area under curve than D rats with hypoglycemic potency of HV > V.

3.3. The mortality of diabetic animals

All animals were monitored for overt signs of vanadium GI stress, Table 2. More than half of V rats developed diarrhea within the first 12 h of treatment, but only one HV rat developed diarrhea.

Table 2
overt signs of vanadium GI stress and the mortality rates

Group label	Overt signs of vanadium GI stress	Mortality rates (%)
D	None	12.5
V	Loss of appetite, Diarrhea, hypoglycemic complication	25
HV	Diarrhea	0
C	None	0

The mortality rate for D rats during the study was 12.5%. For V rats, the mortality rate was 25%. One of the V rats died with a severe hypoglycemic complication (the blood glucose level lower than 2.0 mmol/L, diaphoresis, and depressed) after the first vanadium dosage, and another died with severe diarrhea. In contrast, HV rats showed no death. In addition, no sign of hypoglycemic complication was observed in HV rats.

3.4. The food/water uptake

Food and water intake were recorded and shown in Table 3. The C rats took an average of 78 g of food and 159 mL water per kilogram body weight per day. The D rats consumed an average of 169 g of food and 651 mL of water. There was no significant difference about the average food and water consumption between the vanadium treated rats (V & HV) and the control normal rats. However, the D rats showed a significant higher level of the average food and water consumption than the rats of the test groups.

3.5. The animal weight growth and organ status

Body weights were monitored throughout the study and the results are shown in Fig. 3. The D and C rats gained 0.52, and 1.51 g/day, respectively. The HV rats steadily gained weight until the end of the study with an average rate of about 2.30 g/day, which is higher than that (1.36 g/day) of the V rats.

By the end of the study, about 50% of D rats developed severe cataracta, a common complication of uncontrolled hyperglycemia. There was one case of cataracta in V rats, that also showed vesania, circling and other symptoms. In contrast, no HV rats had cataracta.

3.6. The levels of biochemical markers

Several biochemical indicators of organ toxicity were assessed using ALT as marker of potential liver damage, Table 3. In C and D rats, ALT levels were 113.5 U/L and 136.67 U/L, respectively.

Table 3
Biochemical analyses, the averages of daily water and food consumption in the control normal, diabetic, V-treated and HV-treated diabetic rats

Parameter	C (n = 4)	D (n = 7)	V (n = 6)	HV (n = 8)
Mean blood glucose (mM)	4.43 $\pm$ 0.58	27.46 $\pm$ 4.69	8.66 $\pm$ 4.39*	7.47 $\pm$ 4.12*
Daily mean Water consumption (mL/kg body weight/day)	159 $\pm$ 17	651 $\pm$ 34	172 $\pm$ 17*	150 $\pm$ 17*
Daily mean Food consumption (g/kg body weight/day)	78 $\pm$ 3	169 $\pm$ 6	89 $\pm$ 8*	80 $\pm$ 5*
ALT (U/L)	113.50 $\pm$ 51.49	136.67 $\pm$ 61.67	102.67 $\pm$ 69.59	94.00 $\pm$ 32.63
Urea Nitrogen (mmol/L)	5.03 $\pm$ 0.66	13.17 $\pm$ 4.74	9.30 $\pm$ 2.55	4.68 $\pm$ 2.25*
Creatinine (μ mol/L)	39.00 $\pm$ 3.74	46.67 $\pm$ 6.66	38.37 $\pm$ 4.51	33.60 $\pm$ 7.80
AKP (U/L)	145.75 $\pm$ 16.64	299.33 $\pm$ 49.90	148.00 $\pm$ 46.57*	125.20 $\pm$ 39.20*
LDH (U/L)	2387.00 $\pm$ 758.75	3827.33 $\pm$ 625.35	3616.67 $\pm$ 670.11	1825.80 $\pm$ 136.65*
TC (mmol/L)	1.84 $\pm$ 0.12	1.30 $\pm$ 0.19	1.64 $\pm$ 0.33	1.78 $\pm$ 0.59
TG (mmol/L)	0.49 $\pm$ 0.15	0.41 $\pm$ 0.06	0.72 $\pm$ 0.34	0.59 $\pm$ 0.17
HDL-C (mmol/L)	0.69 $\pm$ 0.08	1.11 $\pm$ 0.21	0.77 $\pm$ 0.23	0.88 $\pm$ 0.18
LDL-C (mmol/L)	0.57 $\pm$ 0.06	0.54 $\pm$ 0.02	0.55 $\pm$ 0.14	0.58 $\pm$ 0.07

Note: The D rats were untreated diabetic, V rats were treated with vanadate-water solution, HV rats were treated with vanadate-herbal decoction, and C rats were non-treatment control normal rats. Data were represented as mean $\pm$ SD.

* $P < 0.05$ compared with untreated diabetic rats (D).

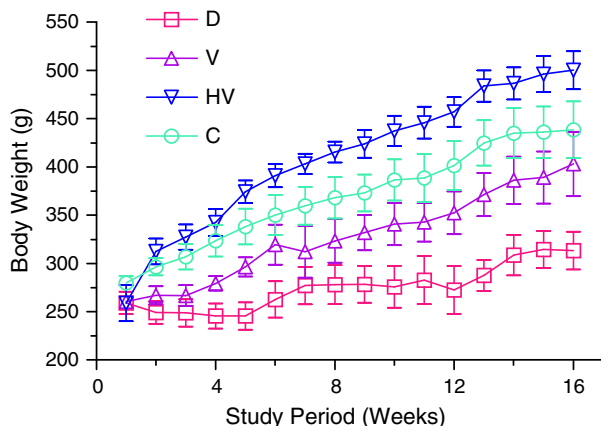


Fig. 3. Mean body weights of diabetic rats during vanadium treatment. Data of the survivors were presented as mean $\pm$ SD with $n = 7, 6, 8$, and 4 for D, V, HV, and C rats, respectively.

The ALT activity was reduced with treatment. Urea nitrogen and creatinine levels, indicators of renal function, are shown in Table 3. The plasma urea nitrogen and creatinine levels increased in the D rats compared with the control. Treatment with vanadate alone reduced the elevated urea nitrogen/creatinine levels, but treatment with vanadate-herbal decoction restored both plasma urea nitrogen and creatinine levels to the normal range, suggesting complete protection of the renal function.

Plasma alkaline phosphatase (AKP) and lactate dehydrogenase (LDH) are measures of potential heart toxicity. The D rats had significantly elevated AKP and LDH levels compared with the control (Table 3). Vanadate-water solution restored the AKP level but had no effect on the LDH levels; however, treatment with vanadate-herbal decoction reduced the LDH and AKP to normal or lower levels. Although certain AKP subfamilies were associated with bone metabolism, we did not assay for AKP subfamily because vanadium accumulation had shown no negative effects on bone (Facchin et al., 2006).

Plasma total cholesterol (TC), triglyceride (TG), high density lipoprotein cholesterol (HDL-C), and low density lipoprotein cholesterol (LDL-C) were also measured (Table 3). Comparing with the control, the D rats showed decreased TC levels and elevated HDL-C levels; however, there were no significant differences in the plasma TG and LDL-C values between the D rats and the other rats. Treatment with vanadate-water solution restored the TC and HDL-C levels, but TG level increased significantly. The vanadate-herbal decoction restored the TC level, partially reduced the TG level induced by vanadate, and kept HDL-C at higher level than the control.

3.7. Vanadium accumulation in animal tissues

The vanadium contents in animal tissues were measured following the studies and the results are shown in Fig. 4. The vanadium content in tissues of C rats was less than $1 \mu\text{g/g}$ (brain, lung, heart, liver, pancreas, spleen, kidney, fat, testis and muscle) while vanadate treatment groups of animals (HV, and V) exhibited elevated levels of vanadium content. Thinking about that the vanadate after reaching the blood stream would be rapidly transported by blood proteins (transferrin and albumin) to various tissues, almost no vanadium stayed in the blood (Mukherjee et al., 2004). Therefore, the vanadium of plasma was not determined. In addition, the vanadium accumulation in bone could improve diabetes-related bone dysfunction, and increase bone formation (Facchin et al., 2006), the vanadium of bone was not determined too. The notable facts include: (1) Ingestion of vanadate with de-

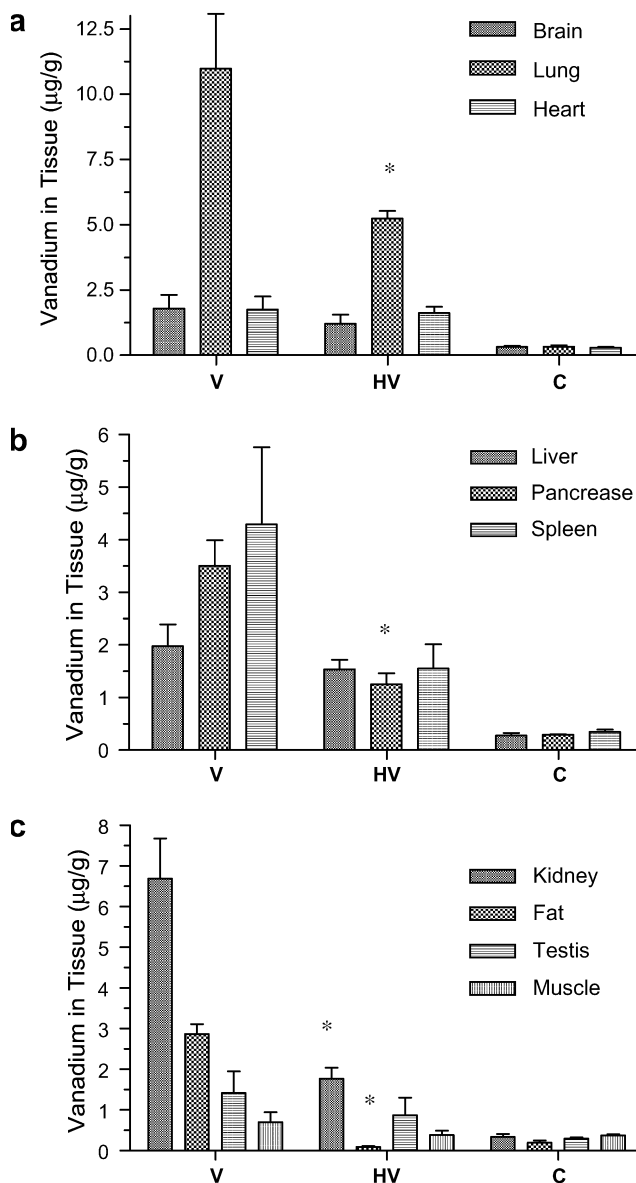


Fig. 4. Levels of vanadium in tissues of diabetic rats upon vanadium treatment. Data of the survivors were presented as mean $\pm$ SD with $n = 7, 6, 8$, and 4 for D, V, HV, and C rats, respectively. * $P < 0.05$ compared with (V) group.

ionized water resulted in high accumulation of vanadium in lung, kidney, spleen and pancreas; (2) Ingestion of vanadate with herbal decoction reduced tissue metal accumulation, particularly for kidney, pancreas, epididymis fat pad, and spleen. The herbal reduction of vanadium accumulation in heart was minor.

4. Discussion

4.1. Ingestion of vanadate with *Salvia miltiorrhiza* Bunge decoction effectively control the glycemic status in STZ rats

The hypoglycemic effects of vanadium compounds were tested in a rat model at a dose of 30 mg/day/animal based on the literature results (Clark et al., 2004a) and our preliminary studies (data not shown). The experimental results showed that ingestion of vanadate with *S. Bunge* decoction effectively controlled the glycemic status, including:

4.3.1. Effective and long-term control of blood glucose level without hypoglycemic shock

During STZ metabolism, various toxic intermediates are produced, including methyl cations, methyl radicals, reactive oxygen species (ROS) and nitric oxide (NO). Beta cells are very susceptible to oxidative changes since they possess a low antioxidative capacity. Injection of STZ induces progressive hyperglycemia, accompanied by lymphocytic infiltration of pancreatic islets (González et al., 2002). As shown in Fig. 1, the untreated D rats had an average glucose level of ~20 mmol/L in the beginning (7 days after STZ injection), but their blood glucose concentration kept rising to 32 mmol/L during the first month, then decreased and stayed at ~27 mmol/L. A growing body of experimental and clinical research indicates that the trace element, vanadium, exerts potent insulin-mimetic effects in vitro and in vivo when used in pharmacological doses (Wilsey et al., 2006). A postulated mechanism of vanadium's insulin-mimetic effects is that vanadium behaves as a phosphate analog and stimulates protein-tyrosine phosphorylation by virtue of its inhibitory actions on phosphatases (PTPase). The administration of vanadate in de-ionized water at a dose of 30 mg/day/animal resulted in rapid decrease of blood glucose levels within 24 h, but the glucose level went up on the second day. The continual treatment of diabetic rats with vanadate-water solution reduced the blood glucose to the control level within one week. Then the average glucose level of 8.6 mM fluctuated and frequently increased so that the regular dose of vanadate was necessary to maintain the control blood glucose level. Interestingly, the blood glucose level of the diabetic rats given the vanadate-herb decoction continually decreased to the control level in ~5 days. Then, the glucose level stayed at an average level of ~7.0 mM for the next two months; an average of 71 ± 3 consecutive days for a blood glucose level of <10 mmol/L. This result suggesting that those antioxidative components of *Salvia miltiorrhiza* Bunge may scavenged free radicals or activated those antioxidant enzyme to reduce the oxidative damage, resulted in those residual beta cells proliferation and function recovery, and enhanced the hypoglycemic effect of vanadate. Yin et al. (2006) have demonstrated that the residual β -cells have the ability to proliferation, although the rate of normal β -cell regeneration is slow. They also observed an important but indirect role for spleen cells in the restoration of endogenous β -cell function. In our study, vanadate alone increased spleen weight. This effect was eliminated with the vanadate-herbal decoction, suggesting that the herbal was protecting spleen functions. Moreover, those antioxidative components of *Salvia miltiorrhiza* Bunge may help to inhibition of β -cell apoptosis. The hypothesis is agree with the report, which American ginseng extracts can suppress UCP-2 in the mitochondria of pancreatic β cells, promoting insulin synthesis and anti-apoptosis (Luo and Luo, 2006). This result suggesting that vanadium-induced amelioration of the diabetic state appears to be secondary to the preservation of a functional portion of the pancreatic β -cells which initially survived STZ-toxicity. This partial preservation of β -cells, although small in proportion to the total insulin store is both critical and sufficient for a chronic reversal of the diabetic state. Thus, such a modest pancreatic preservation can have profound consequences on glucose homeostasis and may underlie the insulin-mimetic effects of vanadium in vivo. More studies are needed.

4.3.2. Improvement of OGTT

As shown in Fig. 2, the combined herbal and vanadium treatment reduced blood glucose levels in the oral glucose administration tolerance test. Ingestion of vanadate in herbal decoction produced the best results. The increase of glucose tolerance may be related to the partial recover of insulin secretion. The observation that HV treatment is capable of enhancing insulin levels and action is interesting. This would provide evidence for a proliferation

of the relic B-cells within the pancreatic islets, which did not been destructed by STZ, preservation of the cells during the progression of the disease, or functional stimulation of existing B-cells. As shown in Fig. 2, vanadate in herbal decoction produced a insulin secretion profile close to the control normal rats.

4.3.3. Improvement of lipid metabolism

As shown in Table 3, the effects of vanadate-herbal on total cholesterol, triglyceride and LDL was a combination of the two entities. The normal size of lighter fat pad was indicative of an overall improvement of lipid metabolism for the vanadate-herbal combination treatment.

4.3.4. Organ protection effect

The absence of tightly regulated blood glucose control in diabetes has led to significant increases in morbidity and mortality (Rosenbloom, 2007). As shown in Table 1, treatment of diabetic rats with vanadate-water solution reduced blood glucose level, but the death rate increased. The mortality rate was comparable with the ingestion of vanadate alone and in combination with herbal decoction. This suggested that (i) the antioxidative protection of the herbal rather than the hypoglycemic effect of vanadium compounds, and (ii) reducing the blood glucose level alone may not be enough to protect a diabetic animal.

Hyperglycemic often made animals to ingest more food and water, and body weight lose. As our experiment, the diabetic rats had limited weight gain with a high rate of food and water consumption. As shown in Fig. 3, vanadate alone enhanced body weight increase, but vanadate in herbal decoction treated rats consumed near normal amount of food and water increasing their body weight to more than the control rats. (Table 3).

In addition to protection from diabetic cataract, based on biochemical markers, the rats ingesting vanadate in herbal decoction exhibited organ protection for liver, heart, and kidney revealed (Table 3). Although the vanadium level in heart was significantly higher than the control, the AKP/LDH clearly suggested a protective effect on the heart.

4.2. Ingestion of vanadate with *Salvia miltiorrhiza* Bunge decoction effectively reduced vanadium toxicity

Vanadium usage as a therapeutic clinical agent has been limited because of its side effects and potential toxicity (Thompson and Orvig, 2006). The ingestion of vanadate with *S. Bunge* herbal decoction apparently eliminated most of these concerns.

First, ingestion of vanadate in herbal decoction greatly reduced the GI stress as indicated by much less diarrhea and near normal body weight and levels of food and water consumption. Although the body weight of rats treated with vanadate-herbal decoction was 10% higher than the normal rats, the epididymis fat pad was at a normal level (data not shown). Second, vanadate alone increased triglyceride level but co-ingestion with herbal decoction moderated this effect. The decreased triglyceride may be related to the high HDL level with administration of herbal-vanadate decoction. Third, co-ingestion significantly reduced tissue vanadium levels, especially for kidney, lung, pancreas, and epididymis fat pad (Fig. 4). The kidney has been considered one of the major organs-1 for vanadium accumulation.

4.3. Summary

The present work has demonstrated that oral administration of vanadate in *S. Bunge* herbal decoction produced a stable (free of hypoglycemic shock) and long-lasting (~70 day) control of glycaemic status in STZ-induced diabetic rats. Effective protection of animal organs from hyperglycemic damage was observed. In addition,

the herb significantly alleviated the side effects of vanadium, i.e. GI stress and metal accumulation. Herbal antioxidants were expected to reduce vanadate metal toxicity.

The study is in overall agree with vanadium compounds as insulin enhancing agents; however, protection/stimulation of pancreatic β -cells could also be involved in the action of vanadium compounds. Nonetheless, it is quite impressive that the herbal decoction produced a synergetic function enhancing the effects of organ protections caused by vanadate, and improved the hyperglycemic effect. More studies are needed, but the present study provides new understanding for therapeutic treatment of diabetics with vanadium compounds.

Conflict of interest statement

The authors declare that there are no conflicts of interest.

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